

Experiment No: 3

Title: Working and Implementation of Platform as a service.

Aim: To Study and implement Platform as a Service using Heroku.

Theory:

Heroku is a cloud platform as a service (PaaS) supporting several programming languages. One of the first cloud platforms, Heroku has been in development since June 2007, when it supported only the Ruby programming language, but now supports Java, Node.js, Scala, Clojure, Python, PHP, and Heroku is said to be a polyglot platform as it has features for a developer to build, run and scale applications in a similar manner across most languages.

Services Provided By Heroku:

The Heroku Platform

The Heroku network runs the customer's apps in virtual containers which execute on a reliable runtime environment. Heroku calls these containers "Dynos". These Dynos can run code written in Node, Ruby, PHP, Go, Scala, Python, Java, or Clojure. Heroku also provides custom buildpacks with which the developer can deploy apps in any other language. Heroku lets the developer scale the app instantly just by either increasing the number of dynos or by changing the type of dyno the app runs in.

Heroku Postgres

Heroku Postgres is the Cloud database (DBaaS) service for Heroku based on PostgreSQL. Heroku Postgres provides features like continuous protection, rollback, and high availability; also forks, followers, and dataclips.

Heroku Redis

Heroku Redis is the customized Redis from Heroku to provide a better developer experience. It is fully managed and is provided as a service by Heroku. It helps in managing instances with a CLI, associate data with Postgres to gain business insights using SQL tools, and lets customer gain performance visibility.

Heroku Teams

Heroku Teams is a team management tool which provides collaboration and controls to bring a customer's developers, processes, and tools together in order to build better software. With Heroku Teams, teams can self-organize, add, and manage members, get fine-grained control with app-level permissions and also use collaboration tools like Heroku Pipelines. It also provides delegated administration and centralized billing.

Heroku Enterprise

Heroku Enterprise provides services to large companies which help them to improve collaboration among different teams. It provides a set of features like fine-grained access controls, identity federation, and private spaces to manage their enterprise application development process, resources, and users.

Heroku Connect

Heroku Connect lets users create Heroku apps that can easily integrate with [Salesforce](#) deployments at scale. This is done by having a seamless data synchronization between Heroku Postgres databases and Salesforce organizations.

Heroku Elements

Heroku Elements provides users with Add-ons (tools and services for developing, extending, and operating the app), Buildpacks (which automate the build processes for the preferred languages and frameworks) and Buttons (a tool for the one-click provisioning, configuring, and deployment of third party components, libraries and patterns).

Advantages:

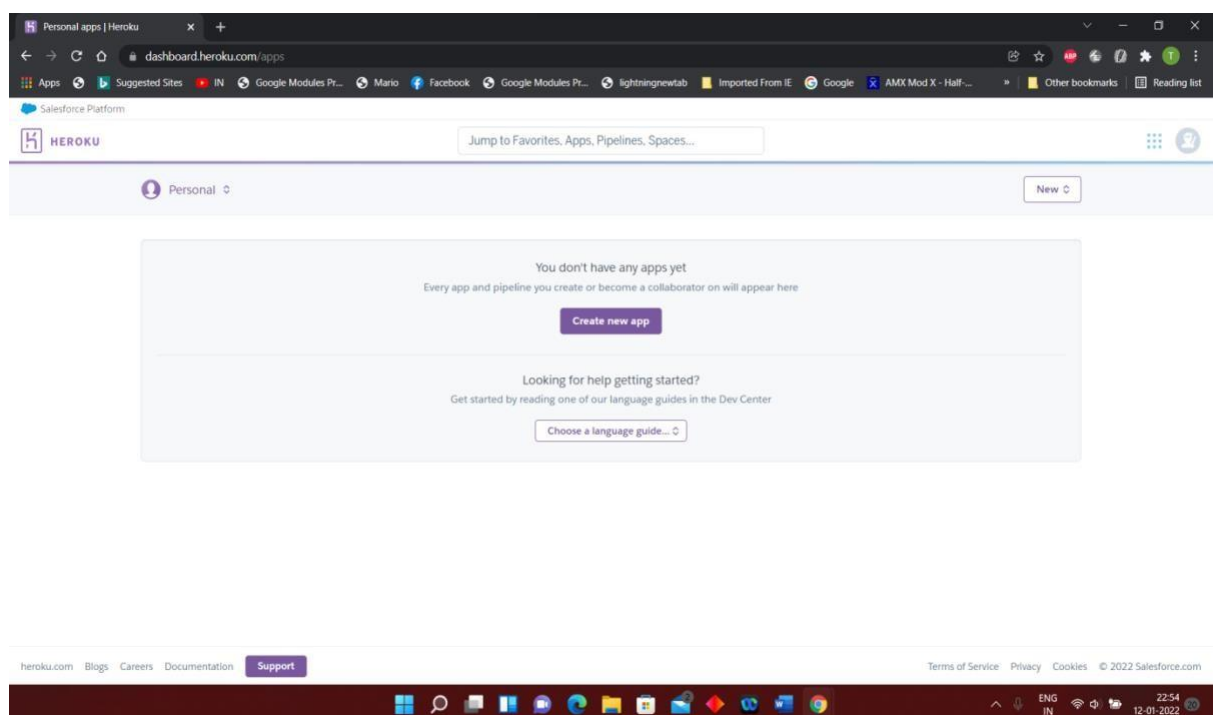
- Easy deployment
- Good for small projects
- Add-ons
- Less maintenance
- Documentation

Disadvantage:

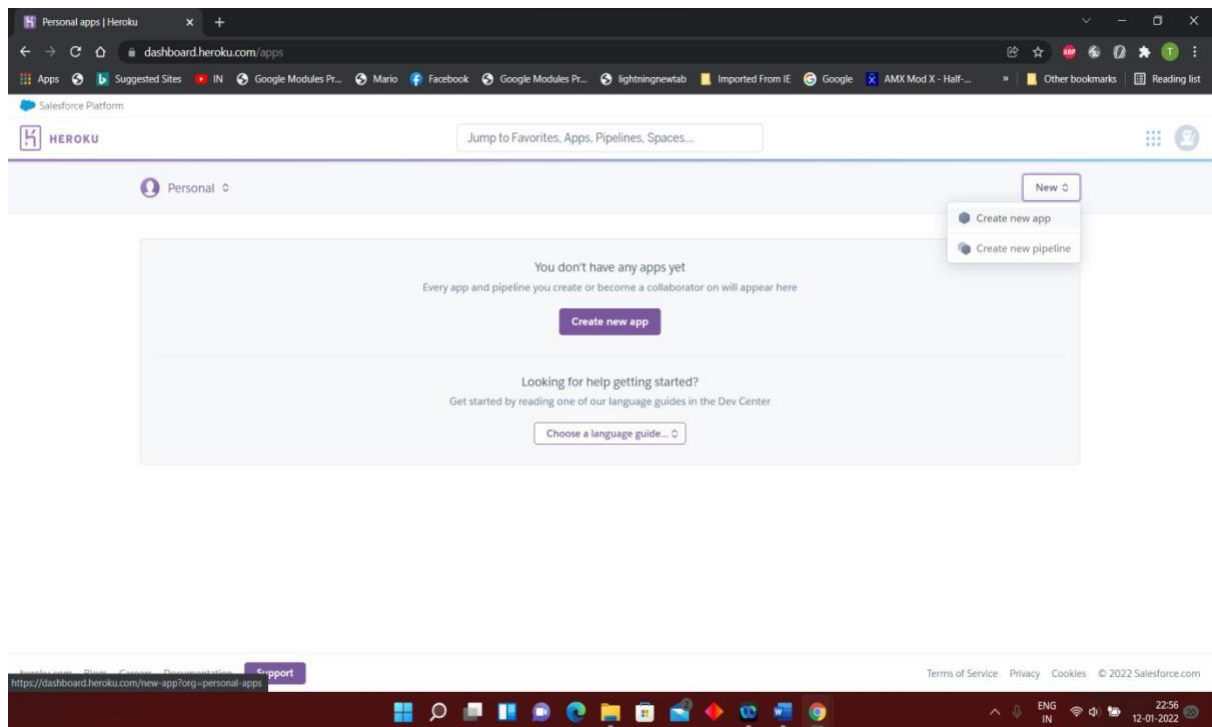
- Price when you need to scale up on your projects

The Steps to Create and Access Platform as a service on Heroku is as follows:

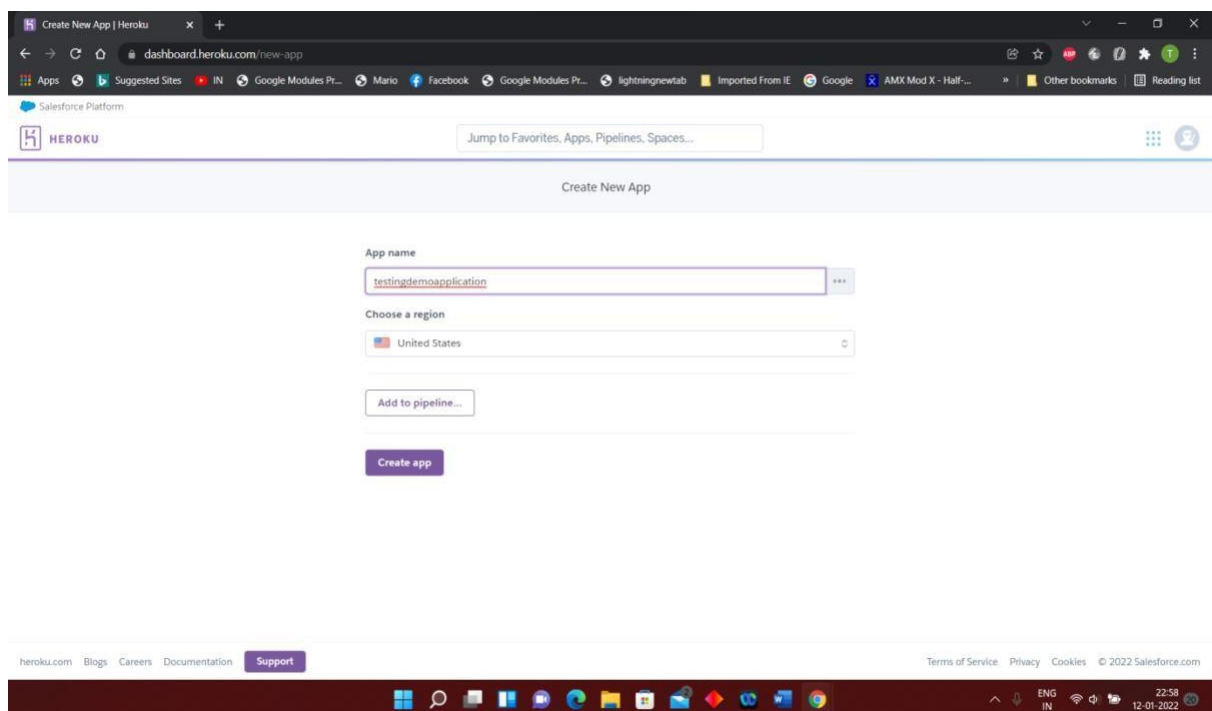
1) Login to Heroku.



2) Click on “New” -> “Create new app” button at top right.



3) Give a name to your app and click “Create app”. Note that this needs to be a unique name.

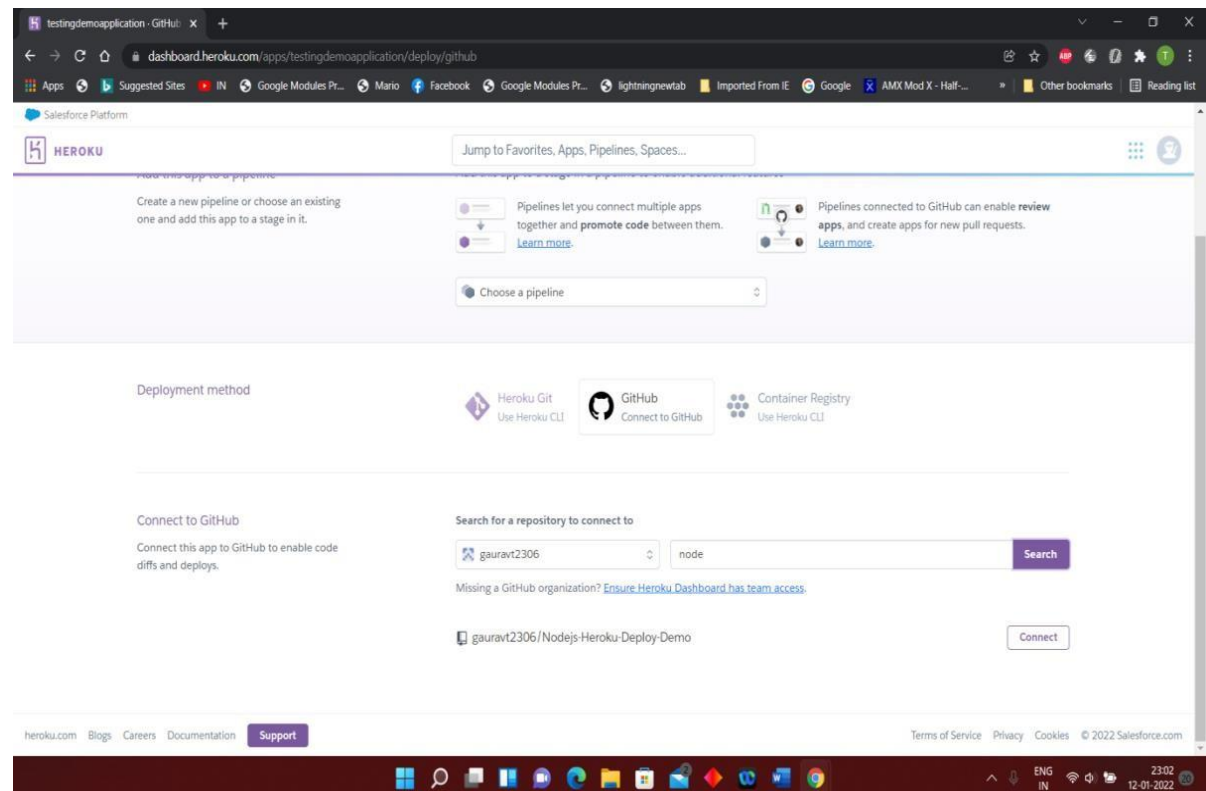


4) You will be redirected to a new page where find the “Deployment method” section and click on Github.

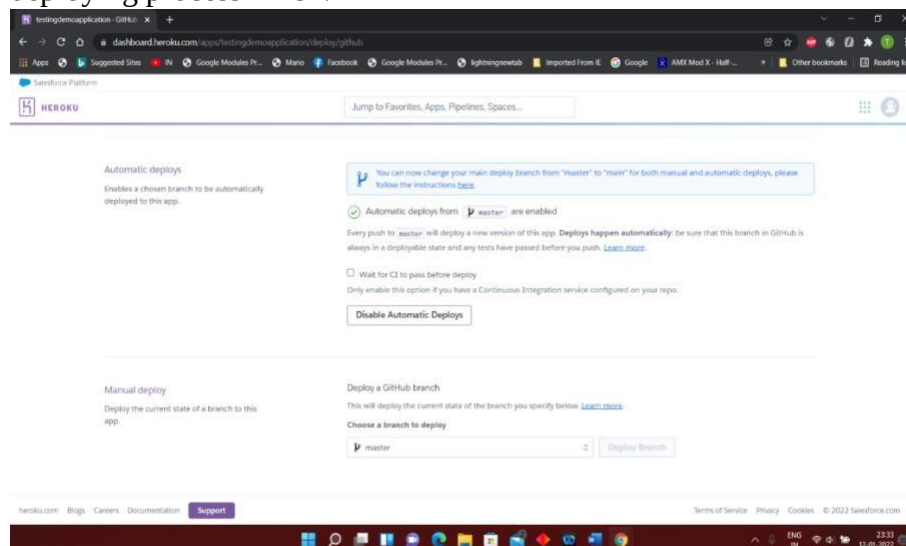
1) You will be redirected to a new page where find the “Deployment method” section and click on Github.

(You can fork this repository for testing: <https://github.com/gauravt2306/Nodejs-Heroku-Deploy-Demo>)

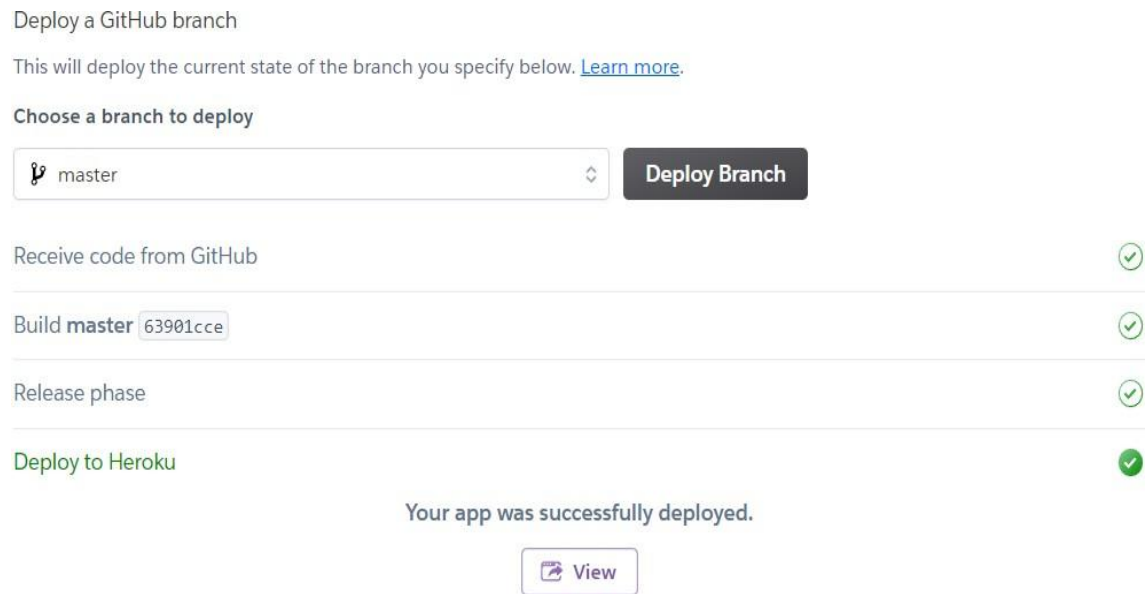
Enter the repository name -> click search -> Select correct repository -> Click connect.



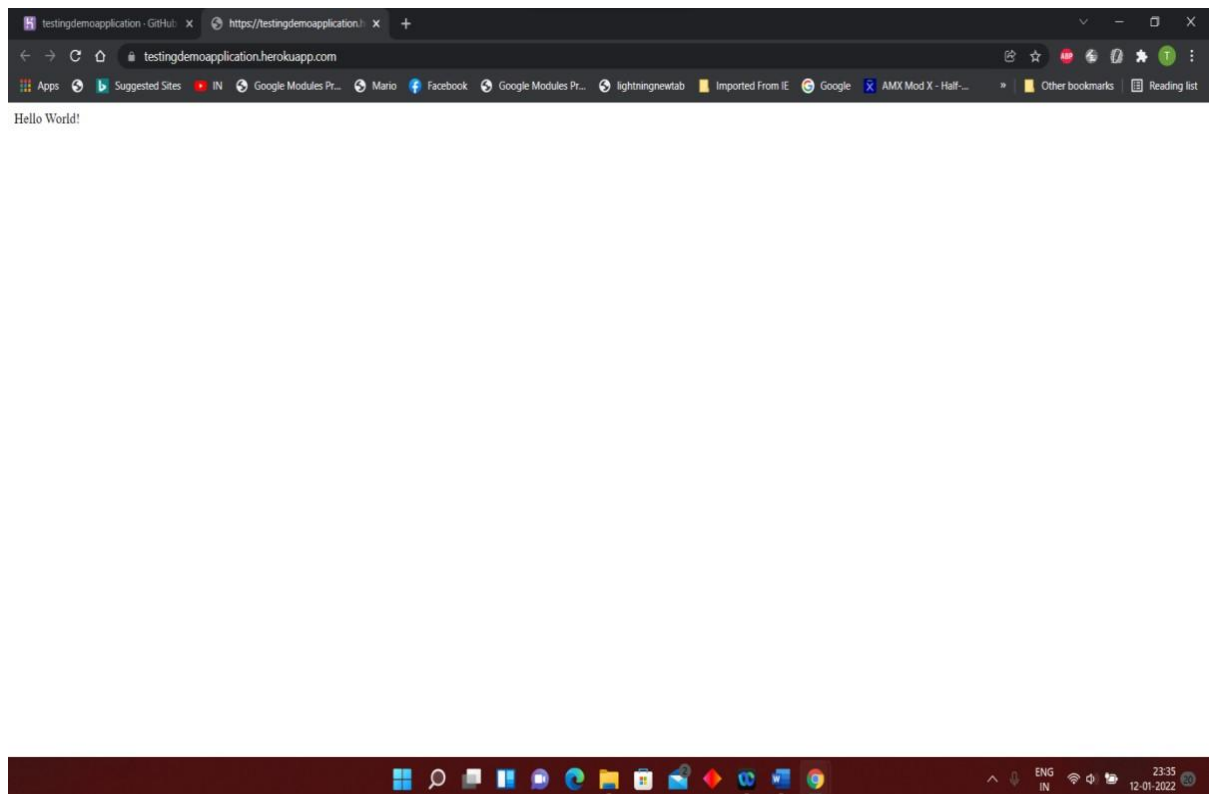
2) Click on “Enable automatic deploys” button which would mean that whenever you add any code to your GitHub repository, Heroku will update your hosted app too. And finally, click on “Deploy branch ”. Wait for a couple of minutes and let the deployment process finish.



3) After successfully deploying the website you will get a success message as shown.



4) Click on View To Open Web Application.



Conclusion: Thus, we have Studied and implemented Platform as a Service using Heroku